VOCO Bangkok Sukhumvit 11 – Table of Column: issued (S-04-01-01 R3) vs ceda-column-design v1.2

Left: cell cropped from the issued drawing. Right: section drawn by the engine for the same loads, f'c, Lu, k as the issued Excel calc (03_xx). D/C = engine 3D-surface check.

Mark	Excel band	Dwg row	Issued bars	D/C issued	Engine bars	D/C engine	ΔAs	Flag
C1	B6-B5	Fdn–B4	66-32	0.98	62-DB32	1.00	-6 %	
C1	B4-B3	B4–B2	44-32 (eval 46)	0.94	42-DB32	0.97	-5 %	
C1	B2-B1	B2–1st	22-32	1.14	30-DB32	1.00	+36 %	issued D/C>1
C1	1st - 3th	1st–4th	22-28	0.98	30-DB25	0.94	+9 %	
C1	4th-10th	4th–11th	44-28 (eval 46)	0.86	30-DB28	0.97	-32 %	
C1	11th-20th	11th–15th	22-25	0.78	30-DB20	0.82	-13 %	
C2	B6-B1	Fdn–B4	44-28 (eval 46)	0.85	30-DB28	0.96	-32 %	
C2	1st-3th	1st–4th	22-28	0.55	30-DB20	0.58	-30 %	
C2	4th-14th	4th–11th	22-25	0.54	30-DB20	0.55	-13 %	
C2s	15th-20th	15th–17th	24-25 (eval 26)	0.76	20-DB25	0.91	-17 %	
C3	B6-B1	Fdn–B4	44-28 (eval 46)	0.85	30-DB28	0.96	-32 %	
C3	1st-3th	Fdn–B4	44-28 (eval 46)	0.64	30-DB20	0.94	-65 %	
C3L	1st-3th_long column	Fdn–B4	44-28 (eval 46)	0.22	30-DB20	0.28	-65 %	
C4	B6-B1	Fdn–B4	44-32 (eval 46)	0.95	42-DB32	0.98	-5 %	
C4L	1st - 3th_long	Fdn–B4	44-32	0.49	32-DB25	0.65	-56 %	
C4	4th-10th	4th–11th	44-28 (eval 46)	0.75	30-DB25	0.93	-46 %	
C4	11th-20th	11th–15th	22-25	0.35	30-DB20	0.36	-13 %	
C5	B6-3th	Fdn–B4	16-28	0.67	16-DB20	0.99	-49 %	
C5L	1st-3th_long	Fdn–B4	16-28	0.37	16-DB20	0.55	-49 %	
C6	4th-10th	4th–11th	60-28	1.02	50-DB32	0.98	+9 %	issued D/C>1
C6	11th-16th	11th–15th	20-28	0.84	30-DB20	0.88	-23 %	
C6	17th-20th	(not on dwg)	—	–	30-DB20	0.85	–	
C7	B6-1st	Fdn–B4	8-28	0.90	8-DB20	0.99	-49 %	
C9	16th-17th	15th–17th	22-25	0.58	28-DB20	0.62	-19 %	
C10	20st	4th–11th	16-28	1.05	16-DB32	0.92	+31 %	issued D/C>1
C12	4th-10th	4th–11th	20-28	0.81	14-DB28	0.98	-30 %	
C12	10th-22th	11th–15th	12-28	0.50	14-DB20	0.63	-40 %	
C12L	10th-20th_long	11th–15th	12-28	0.45	14-DB20	0.56	-40 %	

C1

ISSUED S-04-01-01 R3

ENGINE v1.2 (draft)

Excel band
B6-B5

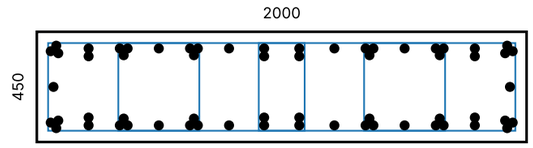
Dwg row
Fdn–B4

f'c 560 ksc
Lu 2.75 m, k 1.0



66 - Ø< 32 mm
3U - Ø<12 mm @ 150 FOR So
3U - Ø<12 mm @ 300 FOR S

issued 66-32 → engine D/C 0.98



62-DB32 bundle 2/3 (incl. non-corner positions) 4 sets DB12 @150/300 D/C 1.00 ρ 5.54 % δ 1.00 OK ΔAs -6 %

Excel band
B4-B3

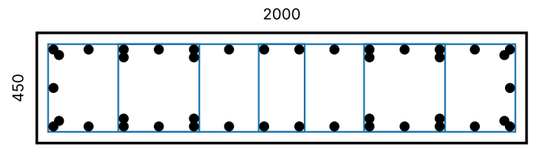
Dwg row
B4–B2

f'c 560 ksc
Lu 4.50 m, k 0.9



44 - Ø< 32 mm
3U - Ø<12 mm @ 150 FOR So
3U - Ø<12 mm @ 300 FOR S

issued 44-32 (eval 46) → engine D/C 0.94



42-DB32 bundle 2 @ tie corners 4 sets DB12 @150/300 D/C 0.97 ρ 3.75 % δ 1.93 OK ΔAs -5 %

Excel band
B2-B1

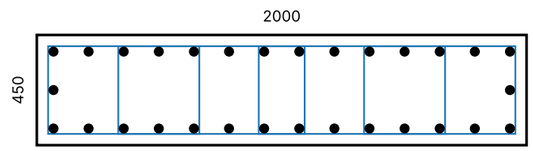
Dwg row
B2–1st

f'c 560 ksc
Lu 4.50 m, k 0.9



22 - Ø< 32 mm
3U - Ø<12 mm @ 150 FOR So
3U - Ø<12 mm @ 300 FOR S

issued 22-32 → engine D/C 1.14



30-DB32 4 sets DB12 @150/300 D/C 1.00 ρ 2.68 % δ 2.07 OK ΔAs +36 %

Excel band
1st - 3th

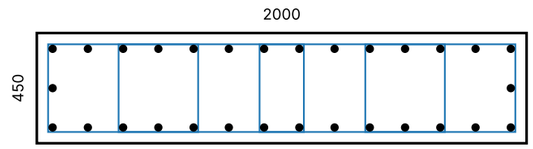
Dwg row
1st–4th

f'c 560 ksc
Lu 4.50 m, k 0.9



22 - Ø< 28 mm
3U - Ø<12 mm @ 150 FOR So
3U - Ø<12 mm @ 300 FOR S

issued 22-28 → engine D/C 0.98



30-DB25 4 sets DB12 @150/300 D/C 0.94 ρ 1.64 % δ 2.18 OK ΔAs +9 %

Excel band
4th-10th

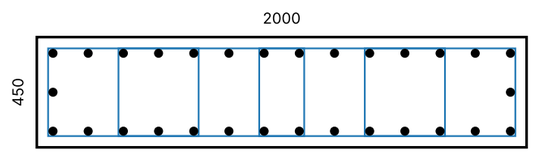
Dwg row
4th–11th

f'c 400 ksc
Lu 4.50 m, k 0.9



44 - Ø< 28 mm
3U - Ø<12 mm @ 150 FOR So
3U - Ø<12 mm @ 300 FOR S

issued 44-28 (eval 46) → engine D/C 0.86

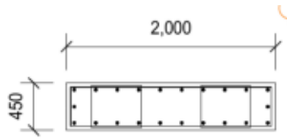


30-DB28 4 sets DB12 @150/300 D/C 0.97 ρ 2.05 % δ 1.86 OK ΔAs -32 %

Excel band
11th-20th

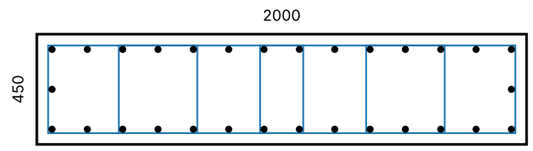
Dwg row
11th–15th

f'c 400 ksc
Lu 4.50 m, k 0.9



22 - Ø< 25 mm
3U - Ø<12 mm @ 150 FOR So
3U - Ø<12 mm @ 300 FOR S

issued 22-25 → engine D/C 0.78



30-DB20 4 sets DB12 @150/300 D/C 0.82 ρ 1.05 % δ 1.59 OK ΔAs -13 %

C2

ISSUED S-04-01-01 R3

ENGINE v1.2 (draft)

Excel band
B6-B1

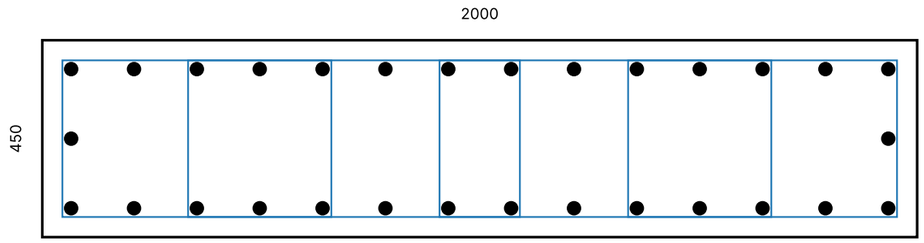
Dwg row
Fdn–B4

f'c 560 ksc
Lu 4.50 m, k 0.9



44 - ∅< 28 mm
3ℓ - ∅<12 mm @ 150 FOR So
3ℓ - ∅<12 mm @ 300 FOR S

issued 44-28 (eval 46) → engine D/C 0.85

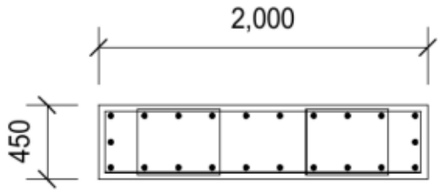


30-DB28 4 sets DB12 @150/300 D/C 0.96 ρ 2.05 % δ 2.11 OK ΔAs -32 %

Excel band
1st-3th

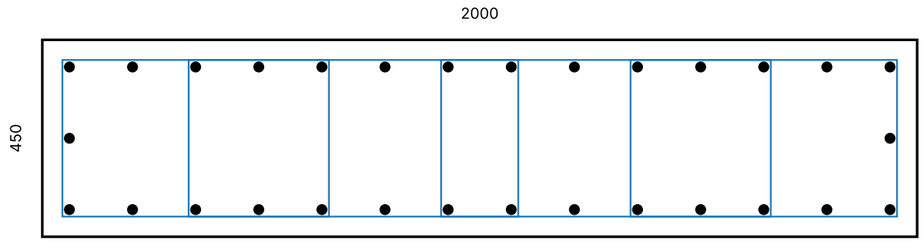
Dwg row
1st–4th

f'c 560 ksc
Lu 4.50 m, k 0.9



22 - ∅< 28 mm
3ℓ - ∅<12 mm @ 150 FOR So
3ℓ - ∅<12 mm @ 300 FOR S

issued 22-28 → engine D/C 0.55



30-DB20 4 sets DB12 @150/300 D/C 0.58 ρ 1.05 % δ 1.65 OK ΔAs -30 %

Excel band
4th-14th

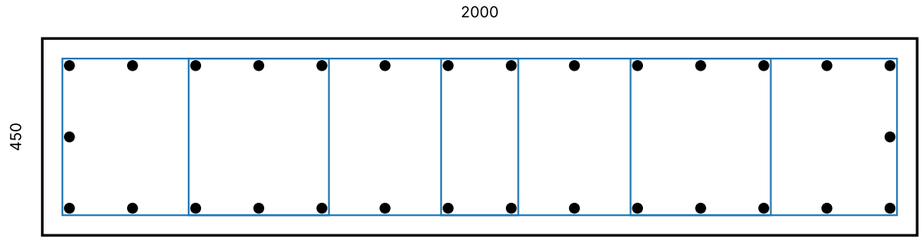
Dwg row
4th–11th

f'c 400 ksc
Lu 4.50 m, k 0.9



22 - ∅< 25 mm
3ℓ - ∅<12 mm @ 150 FOR So
3ℓ - ∅<12 mm @ 300 FOR S

issued 22-25 → engine D/C 0.54

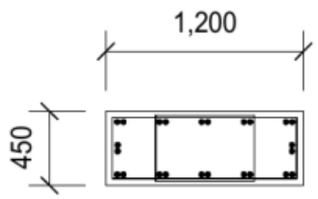


30-DB20 4 sets DB12 @150/300 D/C 0.55 ρ 1.05 % δ 1.46 OK ΔAs -13 %

Excel band
15th-20th

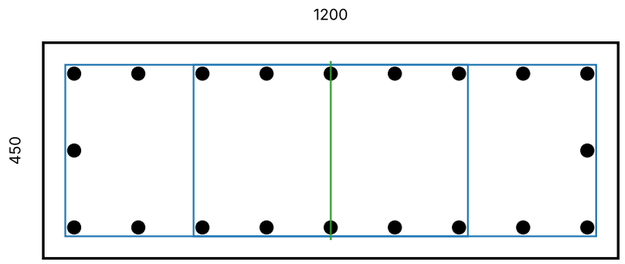
Dwg row
15th–17th

f'c 400 ksc
Lu 4.50 m, k 0.9



24 - ∅< 25 mm
2ℓ - ∅<12 mm @ 150 FOR So
2ℓ - ∅<12 mm @ 300 FOR S

issued 24-25 (eval 26) → engine D/C 0.76

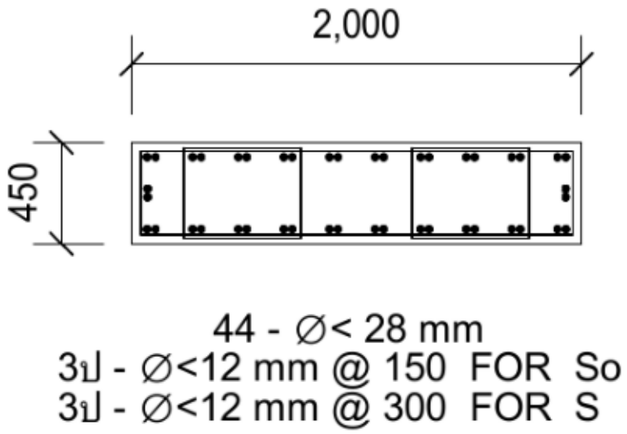


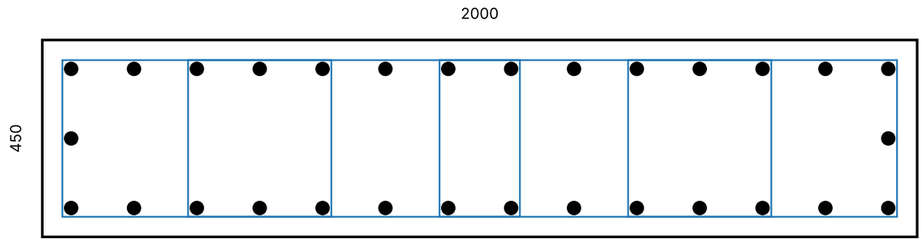
20-DB25 2 sets +1 x-tie DB12 @150/300 D/C 0.91 ρ 1.82 % δ 1.22 OK ΔAs -17 %

Excel band
B6-B1

Dwg row
Fdn–B4

f'c 560 ksc
Lu 4.50 m, k 0.9





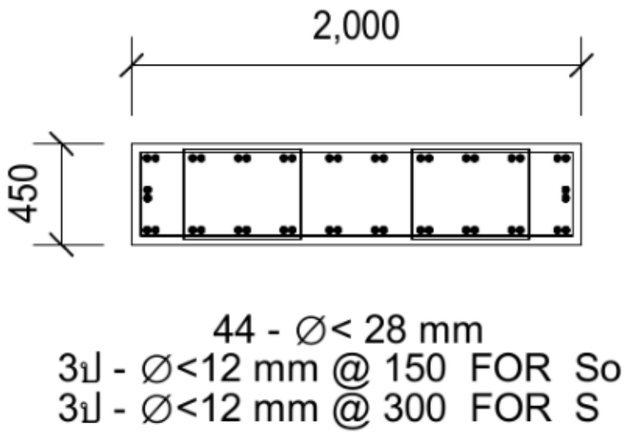
issued 44-28 (eval 46) → engine D/C 0.85

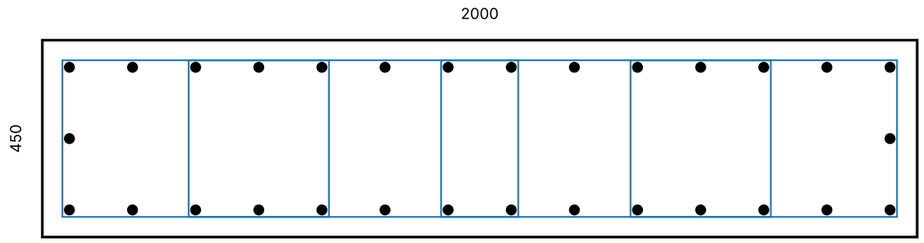
30-DB28 4 sets DB12 @150/300 D/C 0.96 ρ 2.05 % δ 2.11 OK ΔAs -32 %

Excel band
1st-3th

Dwg row
Fdn–B4

f'c 560 ksc
Lu 4.50 m, k 0.9





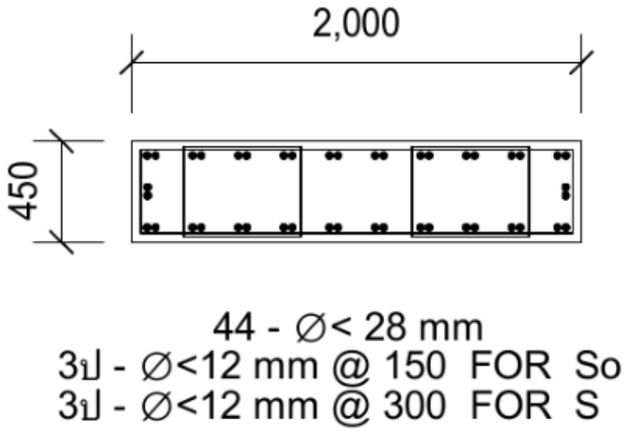
issued 44-28 (eval 46) → engine D/C 0.64

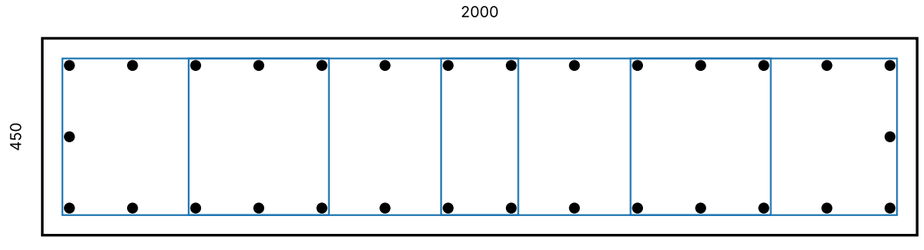
30-DB20 4 sets DB12 @150/300 D/C 0.94 ρ 1.05 % δ 1.92 OK ΔAs -65 %

Excel band
1st-3th_long column

Dwg row
Fdn–B4

f'c 560 ksc
Lu 8.00 m, k 0.9





issued 44-28 (eval 46) → engine D/C 0.22

30-DB20 4 sets DB12 @150/300 D/C 0.28 ρ 1.05 % δ 1.88 OK ΔAs -65 %

C4

ISSUED S-04-01-01 R3

ENGINE v1.2 (draft)

Excel band
B6-B1

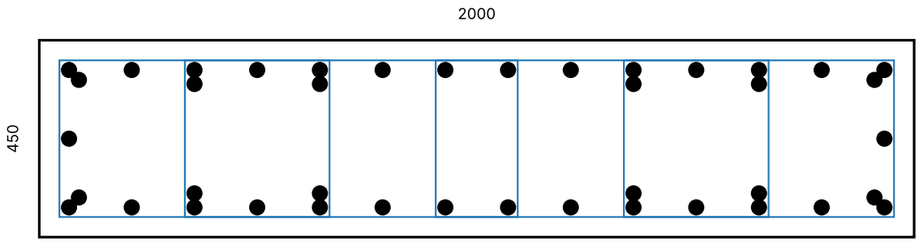
Dwg row
Fdn–B4

f'c 560 ksc
Lu 4.50 m, k 0.9



44 - Ø< 32 mm
3J - Ø<12 mm @ 150 FOR So
3J - Ø<12 mm @ 300 FOR S

issued 44-32 (eval 46) → engine D/C 0.95

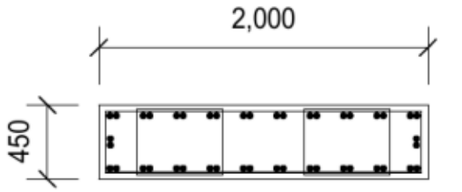


42-DB32 bundle 2 @ tie corners 4 sets DB12 @150/300 D/C 0.98 ρ 3.75 % δ 1.95 OK ΔAs -5 %

Excel band
1st - 3th_long

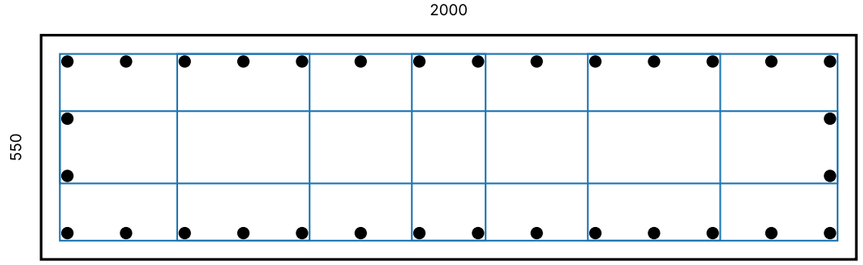
Dwg row
Fdn–B4

f'c 560 ksc
Lu 7.70 m, k 0.9



44 - Ø< 32 mm
3J - Ø<12 mm @ 150 FOR So
3J - Ø<12 mm @ 300 FOR S

issued 44-32 → engine D/C 0.49

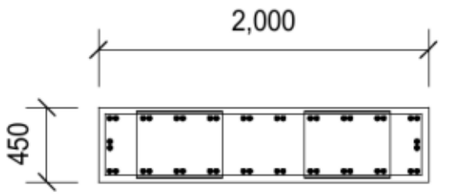


32-DB25 5 sets DB12 @150/300 D/C 0.65 ρ 1.43 % δ 3.00 OK ΔAs -56 %

Excel band
4th-10th

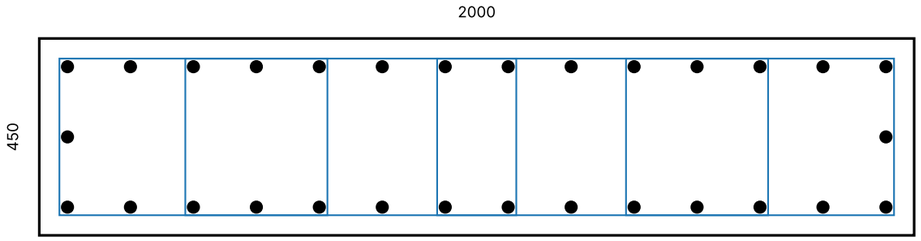
Dwg row
4th–11th

f'c 400 ksc
Lu 4.50 m, k 0.9



44 - Ø< 28 mm
3J - Ø<12 mm @ 150 FOR So
3J - Ø<12 mm @ 300 FOR S

issued 44-28 (eval 46) → engine D/C 0.75

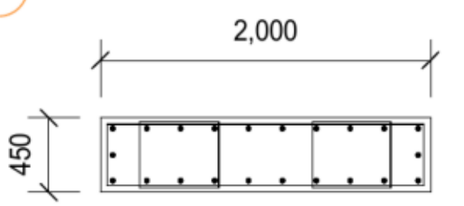


30-DB25 4 sets DB12 @150/300 D/C 0.93 ρ 1.64 % δ 1.57 OK ΔAs -46 %

Excel band
11th-20th

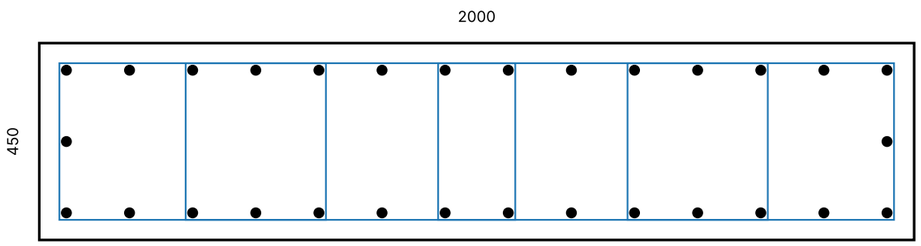
Dwg row
11th–15th

f'c 400 ksc
Lu 4.50 m, k 0.9



22 - Ø< 25 mm
3J - Ø<12 mm @ 150 FOR So
3J - Ø<12 mm @ 300 FOR S

issued 22-25 → engine D/C 0.35



30-DB20 4 sets DB12 @150/300 D/C 0.36 ρ 1.05 % δ 1.27 OK ΔAs -13 %

C5

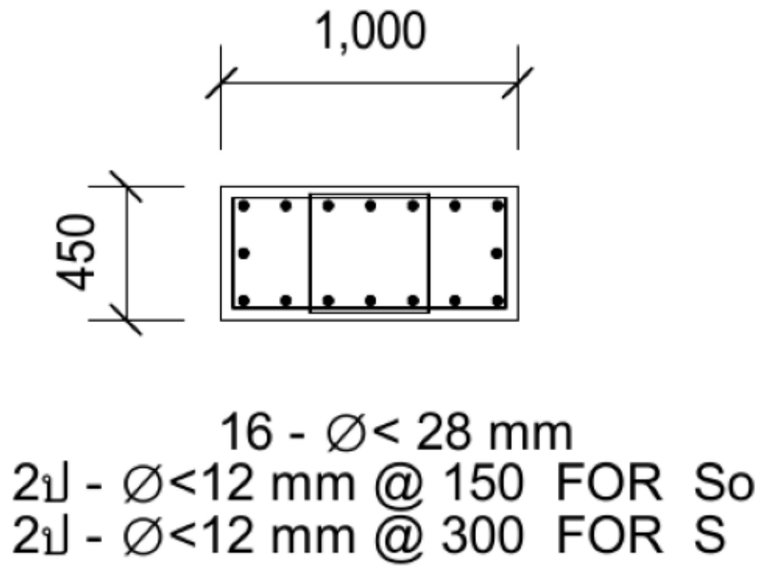
ISSUED S-04-01-01 R3

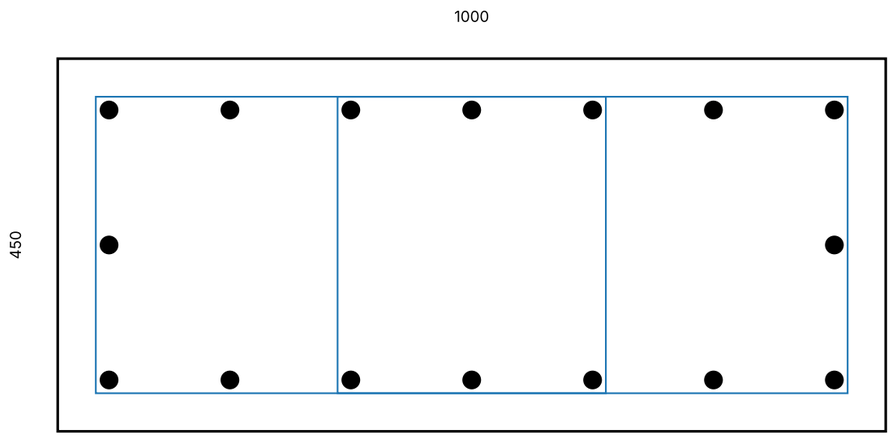
ENGINE v1.2 (draft)

Excel band
B6-3th

Dwg row
Fdn–B4

f'c 560 ksc
Lu 4.90 m, k 0.9





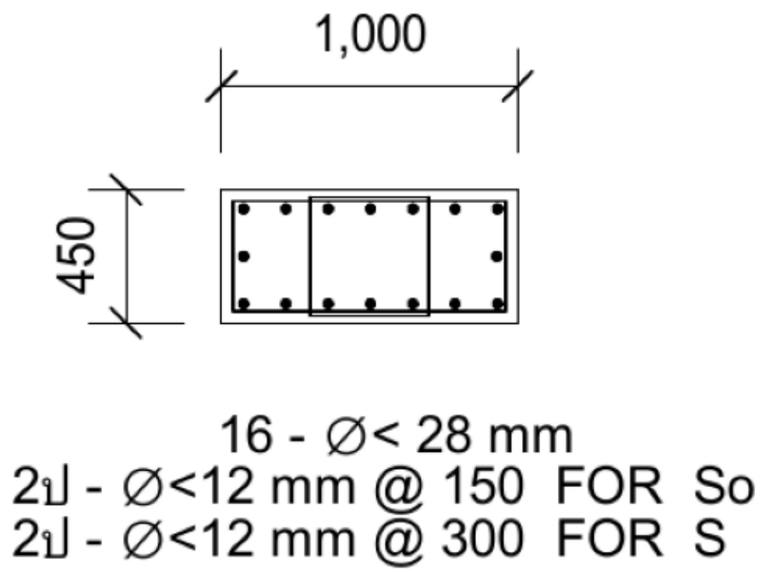
issued 16-28 → engine D/C 0.67

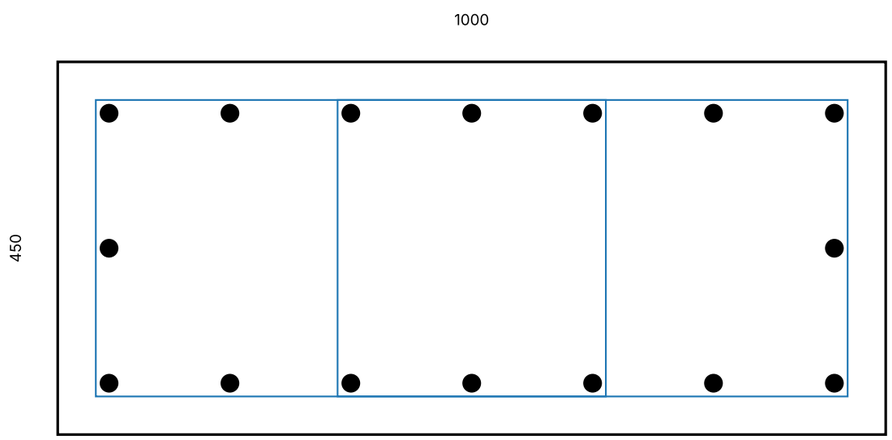
16-DB20 2 sets DB12 @150/300 D/C 0.99 ρ 1.12 % δ 2.34 OK ΔAs -49 %

Excel band
1st-3th_long

Dwg row
Fdn–B4

f'c 560 ksc
Lu 8.00 m, k 0.9





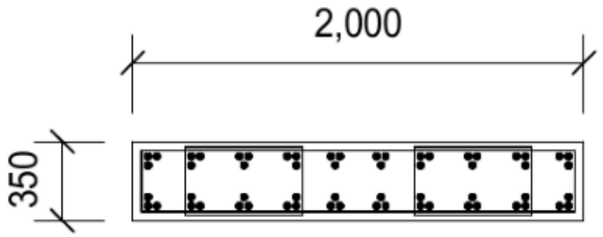
issued 16-28 → engine D/C 0.37

16-DB20 2 sets DB12 @150/300 D/C 0.55 ρ 1.12 % δ 1.62 OK ΔAs -49 %

Excel band
4th-10th

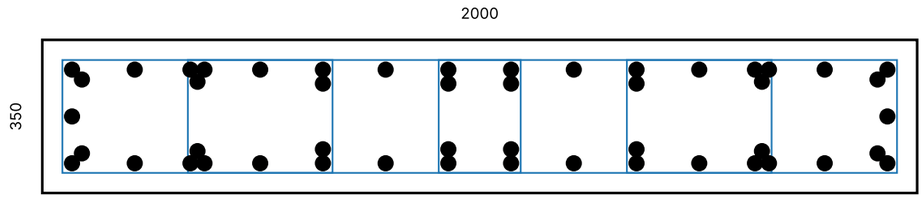
Dwg row
4th–11th

f'c 400 ksc
Lu 3.10 m, k 0.9



60 - ∅< 28 mm
3ป - ∅<12 mm @ 150 FOR So
3ป - ∅<12 mm @ 300 FOR S

issued 60-28 → engine D/C 1.02

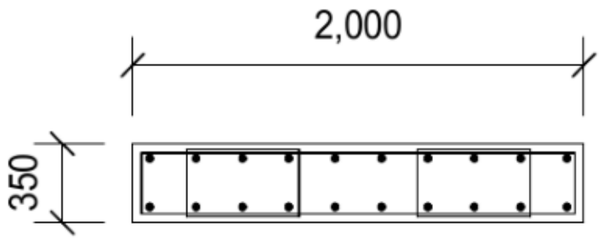


50-DB32 bundle 2/3 @ tie corners 4 sets DB12 @150/300 D/C 0.98 ρ 5.74 % δ 1.53 OK ΔAs +9 %

Excel band
11th-16th

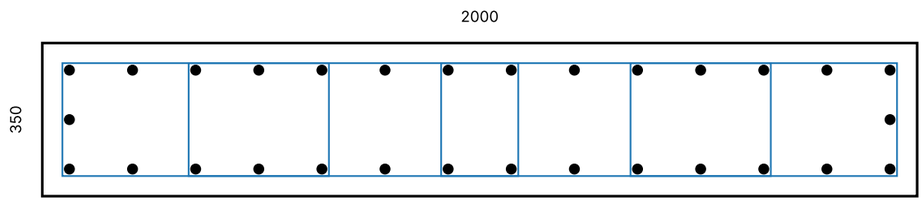
Dwg row
11th–15th

f'c 400 ksc
Lu 3.10 m, k 0.9



20 - ∅< 28 mm
3ป - ∅<12 mm @ 150 FOR So
3ป - ∅<12 mm @ 300 FOR S

issued 20-28 → engine D/C 0.84

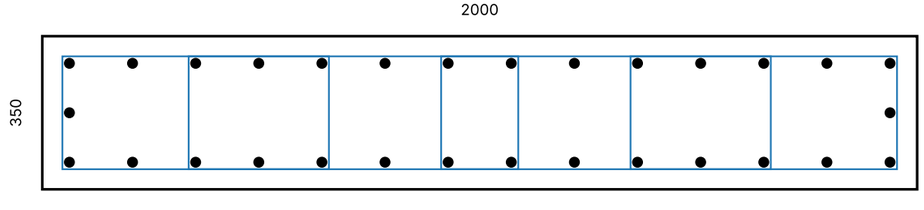


30-DB20 4 sets DB12 @150/300 D/C 0.88 ρ 1.35 % δ 1.64 OK ΔAs -23 %

Excel band
17th-20th

Dwg row
—

f'c 400 ksc
Lu 4.50 m, k 0.9



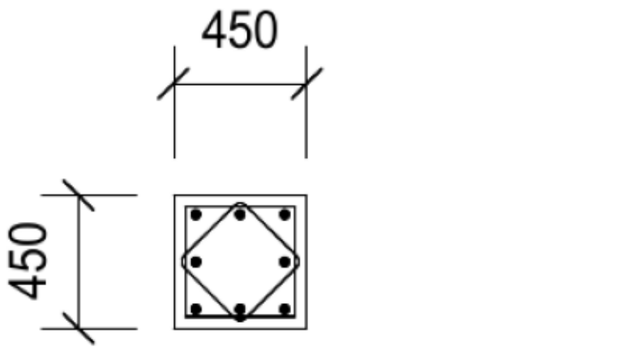
not on drawing

30-DB20 4 sets DB12 @150/300 D/C 0.85 ρ 1.35 % δ 1.91 OK

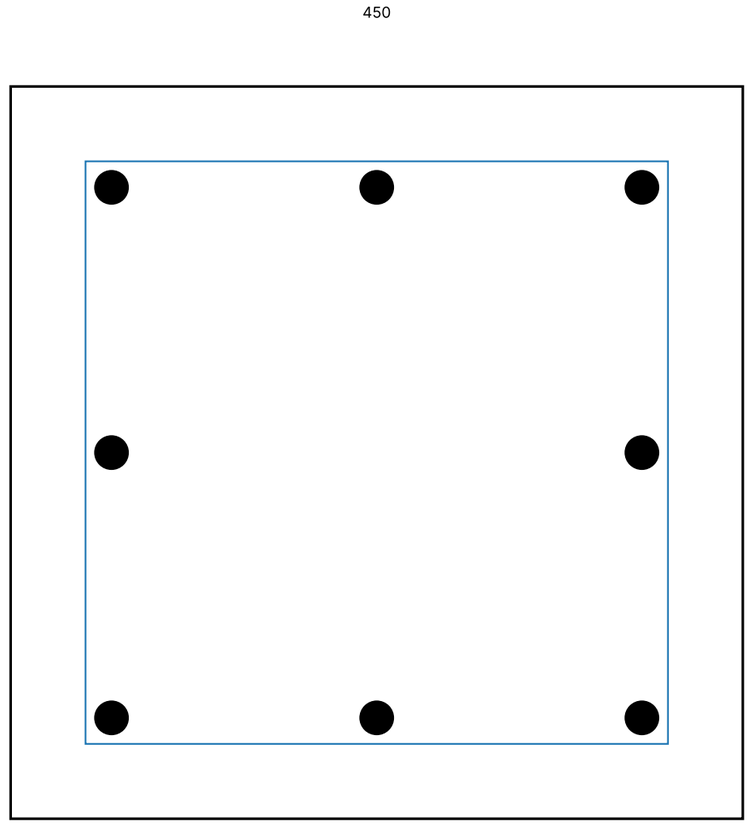
Excel band
B6-1st

Dwg row
Fdn–B4

f'c 560 ksc
Lu 3.00 m, k 0.9



8 - ∅< 28 mm
2ป - ∅<12 mm @ 150 FOR So
2ป - ∅<12 mm @ 300 FOR S



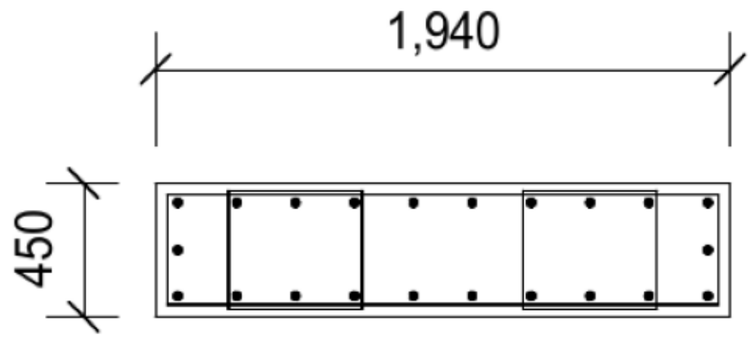
issued 8-28 → engine D/C 0.90

8-DB20 1 sets DB12 @150/300 D/C 0.99 ρ 1.24 % δ 1.00 OK ΔAs -49 %

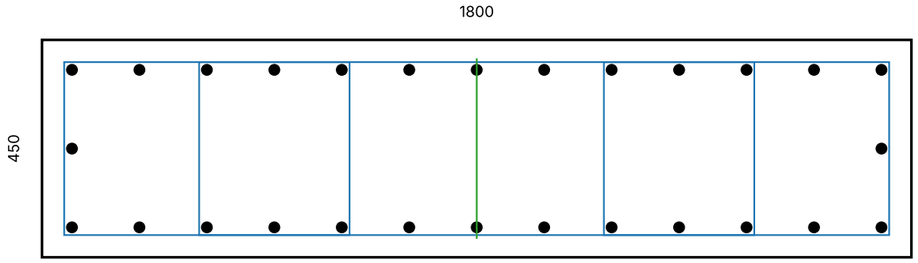
Excel band
16th-17th

Dwg row
15th–17th

f'c 400 ksc
Lu 4.50 m, k 0.9



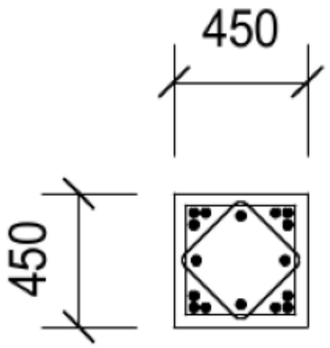
22 - ∅< 25 mm
3ป - ∅<12 mm @ 150 FOR So
3ป - ∅<12 mm @ 300 FOR S



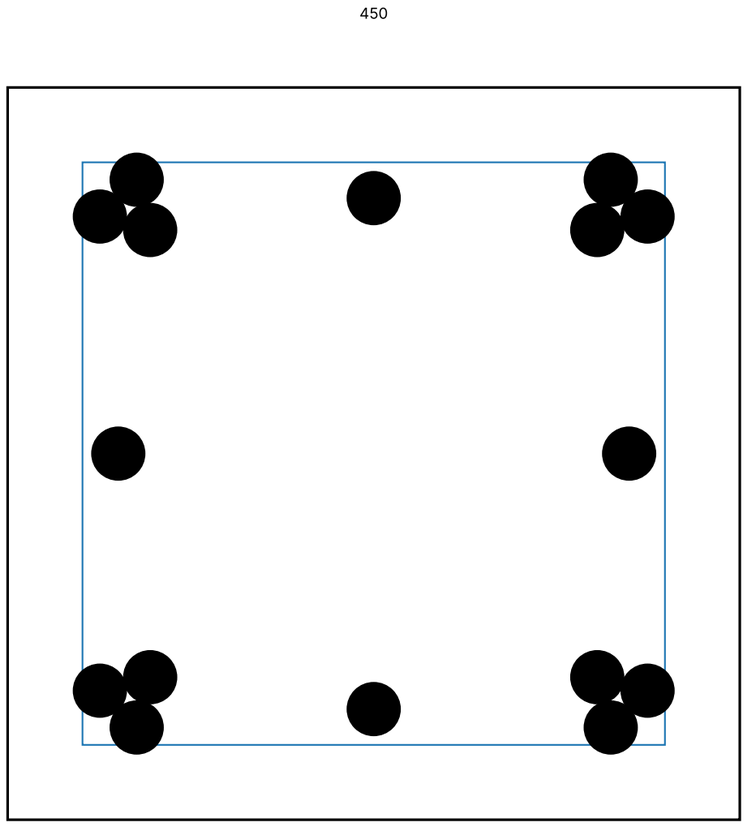
Excel band
20st

Dwg row
4th–11th

f'c 400 ksc
Lu 4.50 m, k 0.9



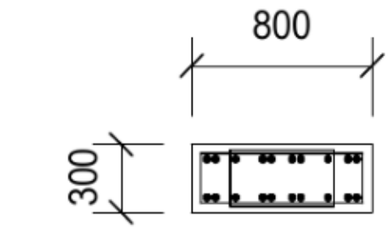
16 - ∅< 28 mm
2ป - ∅<12 mm @ 150 FOR So
2ป - ∅<12 mm @ 300 FOR S



Excel band
4th-10th

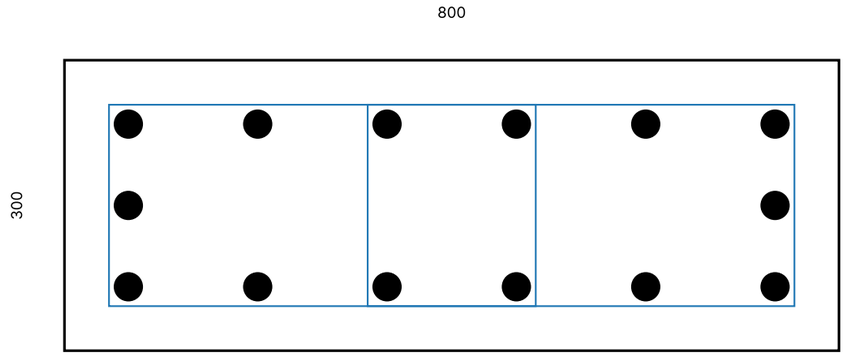
Dwg row
4th–11th

f'c 400 ksc
Lu 3.50 m, k 0.9



20 - ∅< 28 mm
2ปี - ∅<12 mm @ 150 FOR So
2ปี - ∅<12 mm @ 300 FOR S

issued 20-28 → engine D/C 0.81

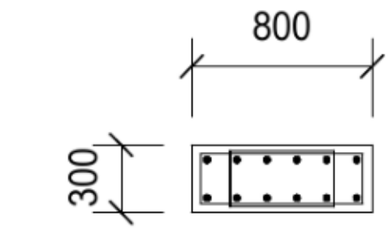


14-DB28 2 sets DB12 @150/300 D/C 0.98 ρ 3.59 % δ 2.28 OK ΔAs -30 %

Excel band
10th-22th

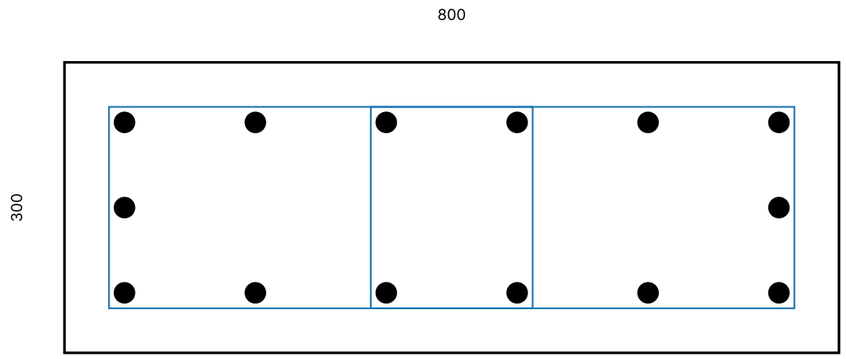
Dwg row
11th–15th

f'c 400 ksc
Lu 3.50 m, k 0.9



12 - ∅< 28 mm
2ปี - ∅<12 mm @ 150 FOR So
2ปี - ∅<12 mm @ 300 FOR S

issued 12-28 → engine D/C 0.50

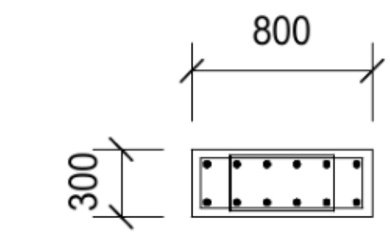


14-DB20 2 sets DB12 @150/300 D/C 0.63 ρ 1.83 % δ 1.68 OK ΔAs -40 %

Excel band
10th-20th_long

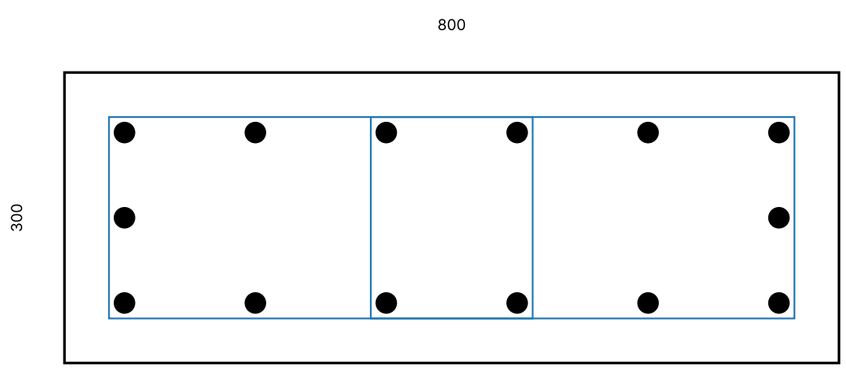
Dwg row
11th–15th

f'c 400 ksc
Lu 4.50 m, k 0.9



12 - ∅< 28 mm
2ปี - ∅<12 mm @ 150 FOR So
2ปี - ∅<12 mm @ 300 FOR S

issued 12-28 → engine D/C 0.45



14-DB20 2 sets DB12 @150/300 D/C 0.56 ρ 1.83 % δ 1.27 OK ΔAs -40 %